Appendix D1 - TCL ‘get’ Command Reference

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get cell Command

Command	Expression/Target object	Options	Additional Options	Allowed Arguments	Description	Overlap with existing commands
get cell	**<pattern/name>**	-regexp/-glob/-exact	?-type?	mem	get cell *cell_* -glob -type mem : returns cells having string “cell_” and type memory, in the library	
				switch	get cell (cell_) -regexp -type switch : returns cells having pattern <cell_> and type switch, in the library	
				io	get cell * -glob -type io : returns all cells of type io, in the library	
				clock_inst	get cell * -glob -type clock_inst : returns all cells whose instances are part of clock-network	
				flop	get cell * -glob -type flop : returns all cells of type flop in the library	
				latch	get cell * -glob -type latch : returns all cells of type latch in the library	
				macro	get cell * -glob -type macro : returns all cells of type macro	
				decap	get cell decap1 -exact -type decap : returns “decap1” if there is a cell of type decap in the library, else returns error	
		-glob	-bbox	-bbox <llx><lly> <urx><ury>	get cell * -glob -bbox <llx> <lly> <urx> <ury> returns all cell masters instantiated inside the specified bounding box. Note: -type is not supported with the –bbox option.	
	<target cell-name>	-type			get cell cell1 -type : returns the celltype of cell1 (e.g., return value: primitive combinational)	get celltype
		-height			get cell cell1 -height : returns height of cell1	query inst -bbox
		-width			get cell cell1 -width: returns width of cell1	
		-pins	?-direction?	input	get cell cell1 -pins -direction input : returns input pins of cell1	
				output	get cell cell1 -pins -direction output : returns output pins of cell1	
				inout	get cell cell1 -pins -direction inout : returns inout pins of cell1	
			?-type?	power	get cell cell1 -pins -type power : returns power-pins of cell1	
				clock	get cell cell1 -pins -type clock : returns clock-pins of cell1	
				signal	get cell cell1 -pins -type signal : returns signal-pins of cell1	
				ground	get cell cell1 -pins -type ground : returns ground-pins of cell1	
		-instcount			get cell cell1 -instcount : returns number of instantiations for cell1	
		-pgarcs			get cell cell1 -pgarcs : returns pgarc(s) of cell1 as TCL list (e.g., return value: VDD VSS VDDC VSS)	

get inst Command

Command	Expression/ Target object	Options	Additional Options	Allowed Arguments	Description	Overlap - existing commands
get inst	<pattern/ name>	-regexp/ -glob/- exact	? -type? ? -exclude ?	mem	get inst *inst_* -glob -type memory : returns all instances having string “inst_” and celltype memory. Or excludes memory cell types with -exclude.	get inst
				switch	get inst (inst_) -regexp -type switch : returns all instances in the design having pattern <inst_> and celltype switch	
				io	get inst * -glob -type io : returns all instances in the design of celltype io. Or excludes io cell types with -exclude.	
				clock_inst	get inst * -glob -type clock_inst : returns all instances that are in the clock network. Or excludes clock_inst types with -exclude.	
				flop	get inst * -glob -type flop : returns all instances in the design of celltype flop. Or excludes flop cell types with -exclude.	
				latch	get inst * -glob -type latch : returns all instances in the design of celltype latch. Or excludes latch cell types with -exclude.	
				leaf	get inst * -glob -type leaf : returns all leaf-instances in the design. Or excludes leaf instance types with -exclude.	
				block	get inst * -glob -type block : returns all block-instances in the design. Or excludes block instance types with -exclude.	
				decap	get inst inst1 -exact -type decap : returns “inst1” if there is a decap-instance in design with name “inst1” , or returns error	
				clock	get inst * glob -exclude [clock | switch] returns a list of instances that excludes clocks or switches	
				std_cells, combinational	Returns a list of standard cells or combinational cells in the design.	
			?-power -type all?	-pin <pin_name>	Returns the power for a particular arc of an instance	
			?-nearest_switch		get inst inst1 -nearest_switch: returns the switch physically nearest to inst1	
			?-master?	<cellname>	get inst * -glob -master cell1 : returns all instances of cell1	get instofcell
			?-location ?	<x> <y>	get inst * -glob -location <x> <y> : returns a list of instances that are present at the specified point	
			?-net?	<netname>	get inst * -glob -net net1 : returns all instances connected to net1	get instbynet
			?-layer?	<layername>	get inst * -glob -layer layerAB: returns all instances on layerAB. Specifying ‘-o <output file>’ avoids potential very high memory use.	
			?-switching?	?<netname>? ?-out_file <fname>?	get inst * -glob -switching : returns all switching instances. valid after dynamic simulation. get inst * -glob -switching -net <netname>: returns a list of switching instances connected to specified net.	print sw
			?-bbox?	<x1 y1 x2 y2>	get inst * -glob -bbox x1 y1 x2 y2 : returns all instances placed inside the bbox (x1 y1 x2 y2)	
			-master_list	{mastercell1 mastercell2 ... } -out_file <fname>	get inst * -glob -master_list { cell1 cell2 }: returns all instances of cells cell1 or cell2	
			? -pin <pin_name>?	-layer -location all_points	get inst inst1 -pin VDD -layer: returns the layers of the specified pin VDD of inst1 get inst <instName> -pin <pinName> -location all_points: returns the location and layer of all geometries of the specified pin.+	
			-ipf_coverage		get inst <pattern> -ipf_coverage: returns ipf coverage status of specified instance -- “1” if ipf coverage is provided, “0” if not.	
			-vcd_coverage		get inst <pattern> -vcd_coverage: returns vcd coverage status of specified instance -- “1” if vcd coverage is provided, “0” if not.	
			-sta_coverage	[output_pin_tw_coverage | clock_pin_tw_coverage | clock_pin_tw_gt_output_tw | clk_pin_not_connected | mintw_gt_maxtw | sta_file_coverage]	get inst <pattern> -sta_coverage <coverage type>, returns STA coverage status for output pins of specified instance and coverage type. For example, for “clk_pin_not_connected”, returns “1” if this is true in the STA file for the instance; otherwise returns “0”.	
			-spef_coverage	[loadless| driverless| driver_mismatch | invalid_nets]	get inst <pattern> -spef_coverage loadless, returns the coverage status of output load type specified in spef. For example, for “loadless”, returns “1” if the output load is not specified in the SPEF file for the instance; otherwise returns “0”.	
			-load		get inst <pattern> -load: returns the load value of the instance	
			-slew		get inst <pattern> -slew : returns slew value used for the instance	

			-freq		get inst <pattern> -freq: returns the frequency used for the instance.	
			-clock_domain		get inst <pattern> -clock_domain: returns the clock index of the instance in the STA file.	
			Note: *-net and –type options do not work together, –net and –master options do not work together, and –net and –bbox options do not work together.*			
	<Targ inst-name>	-master		?(-out_file <fname>?	get inst inst1 -master : returns master-cell name of inst1	get master
		-orientation			get inst inst1 -orientation : returns the orientation of inst1 (e.g., return values: north, fwest, feast, unknown)	get orientation
		-location			get inst inst1 -location : returns the LL coordinate of inst1	query inst -bbox
		-coords			get inst inst1 -coords : returns all coordinates of a bbox or polygon	
		-peak_current	?-pin?	<pgpinname>	get inst inst1 -peak_current -pin VDD : returns peak-current for VDD pin at inst1	
		-frequency			get inst inst1 -frequency : returns the frequency of inst1	print inst_data
		-toggle_rate			get inst inst1 -toggle_rate: returns the toggle rate of inst1	print inst_data
		-bbox			get inst inst1 -bbox, returns: LL and UR x,y coordinates of inst1	
		-charge			get inst inst1 -charge, returns the calculated charge for the specified instance.	
					get inst inst1 -load, returns the capacitive load for the specified instance.	
		-power {supports hierarchical block}	?-type?	leakage	get inst inst1 -power -type leakage : returns the leakage power for inst1	print inst_data
				internal	get inst inst1 -power -type internal : returns the internal powers for inst1 for all power-pins	
				switching	get inst inst1 -power -type switching : returns the switching powers for inst1 for all power-pins	
				all	get inst inst1 -power -type all : returns avg powers for inst1 for all power-pins	
			-type all	-pin <pin_name>	Returns the power for a particular arc of an instance	
		-pins			Returns a list of the names of all the pins of an instance	
		-connectivity			Returns a list of all nets connected to an instance.	
		-voltage -voltage	?-pgarc? -drop	<pgarc name as list>	get inst inst1 -voltage -pgarc VDDC VSS : returns voltage-drop between VDDC-VSS instance P/G pins	
					get inst inst1 -voltage -drop : returns the voltage drop on the instance	
				all	get inst inst1 -voltage -pgarc all : returns voltage-drop of all pgarcs of inst1, specified by P/G pins	
			?-arc?	<arc name as TCL list>	get inst inst1 -voltage -arc VDDC VSS : returns voltage-drop between VDDC-VSS net-domains	
			?-type?	static_ir ? -pin <name>?	get inst inst1 -type static_ir : returns worst static ir-drop for inst1. Optionally can include voltage by pin name.	
			?-dvd_type?	minWC	get inst inst1 -voltage -dvd_type minWC : returns minWC voltage of inst1. Valid after dynamic sim. Can be used with -arc|-pgarc option.	print inst_data
				minTW	get inst inst1 -voltage -dvd_type minTW : returns minTW voltage of inst1. Valid if inst1 has TW and after dynamic sim. Can be used along-with -arc|-pgarc option.	
				maxTW	get inst inst1 -voltage -dvd_type maxTW : returns maxTW voltage of inst1. valid if inst1 has TW and after dynamic simulation. Can be used along-with -arc|-pgarc option.	
				avgTW	get inst inst1 -voltage -dvd_type avgTW : returns avgTW voltage of inst1. valid if inst1 has TW and after dynamic simulation. Can be used along-with -arc|-pgarc option.	
		-nearest_pad	-pin <pin_name> -net <net_name>	-type [power|ground]	get inst inst1 -nearest_pad: returns the pad physically nearest to each power and ground pin of inst1	
		-switching_status			get inst inst1 -switching_status: returns the switching status (Switching/ Not switching) of inst1. valid after dynamic simulation.	print inst_data
		-resistance	-net <net_name>		Returns the minimum path resistance of the instance and net to the nearest pad.	
			-pin <pin_name>	-res_type [avg|min|max] -location all_points	Returns the min, avg, or max path resistance of the specified pin to the nearest pad. Default is "avg". Returns the location and the layer name of all geometries of the specified instance pin.	

		-timing_window	-type	rise fall used	get inst inst1 -timing_window -type rise: returns the maximum and minimum rise time of the first pin of inst1. get inst inst1 -timing_window -type fall: returns the maximum and minimum fall time of the first pin of inst1. get inst inst1 -timing_window -type used: returns the timing window of inst1. For instances not covered in STA, reports the default timing window.	

get net Command

Command	Expression/Target object	Options	Additional Options			Allowed Arguments
get net	**<pattern/name>**	-glob/-regexp/-exact	?-type?	power	get net *VDD* -glob -type power : returns all power nets in the design, having string "VDD"	
				signal	get net (sig_) -regexp -type signal : returns all signal nets in the design, having pattern (sig_)	
				ground	get net VSS -exact -type ground : returns VSS if VSS is a ground net in the design, else return error	
				clock	get net * -glob -type clock : returns all clock nets in the design	
			?-gsr_only?		get net * -glob -type power|ground -gsr_only : returns power|ground nets defined in gsr. Works along with -type option only.	
			?-inst?	<inst_name>	get net * -glob -inst inst1 : returns all nets connected with inst1	
			?-pin?	<pin_name>	get net * -glob -pin VDD : returns the net connected with VDD pin of top instance. get net * -glob -inst sw1 -pin **VDD_EXT : returns the net connected with VDD_EXT pin of instance sw1**	
			-bbox	-bbox <llx> <lly><urx><ury>	get net * -glob -bbox <llx> <lly> <urx> <ury> returns all nets inside the specified bounding box. Note: -type is not supported with the –bbox option.	
	<Target net-name>	-ideal_voltage			get net VDD -ideal_voltage : returns ideal voltage of net VDD	
		-worst_drop			get net net1 -worst_drop : returns the worst drop on net1. valid after static/dynamic analysis	
		-worst_em			get net net1 -worst_em : returns the worst em % on net1. valid after static/dynamic analysis	
		-connectivity			Returns a list of all nets connected to the net.	
			-current -mode	[avg| peak| rms]	Returns selected type of current for signal EM analysis. Default is average current.	
		-connectivity			returns all nets connected to the specified net	

get pad Command

Command	Expression/Target object	Options	Additional Options	Allowed Arguments	Description	Overlap with existing commands
get pad	**<pattern/name>**	-glob/-regexp/-exact	?-type?	power| ground	**Note: 'get pad' can be performed prior to extraction**. get pad *VDD/VSS* -glob -type power /ground: returns all pwr/gnd pads in the design with pattern VDD or VSS	
			?-net?	<net_name>	get pad pad1 -glob -net abc returns all pads associated with net abc	
			-internal net	?<output_file>?	get pad * -glob -internal_net: returns a list of all internal pads only	

	<Target pad-name>	-layer			get pad pad1 -layer : returns the metal layer of pad1	get pad -info
		-location			get pad pad1 -location : returns the LL coordinate of pad1	get pad -info
		-current			get pad pad1 -current : returns current (peak for dynamic) for pad1	

get design Command

Command	Options	Additional Options	Allowed Arguments	Description	Overlap with existing commands
get design				get design : returns the name of design	
	-clock_roots			get design -clock_roots : provides a list of clocks in the design based on the information from the clock section of the STA file.	
	-bbox			get design -bbox : returns the boundary-box (LL and UR coordinates) of design	
	-worst_inst	?-dvd_type?	minWC	get design -worst_inst -dvd_type minWC : returns the instance having worst minWC drop in the design. "get design -worst_inst …" valid after static/dynamic analysis	
			minTW	get design -worst_inst -dvd_type minTW : returns the instance having worst minTW drop in the design	
			maxTW	get design -worst_inst -dvd_type maxTW : returns the instance having worst minTW drop in the design	
			avgTW	get design -worst_inst -dvd_type avgTW : returns the instance having worst avgTW drop in the design	
		-static		get design -worst_inst -static : returns the instance name with the worst voltage, and its voltage in Static IR analysis	
	worst_inst_drop	-static	-limit <N>	get design -worst_inst_drop -static : returns worst-inst name and worst static voltage-drop. -limit <N> : displays the top N violations	
		?-dvd_type?	minWC -limit <N>	get design -worst_inst_drop -dvd_type minWC : returns the worst minWC drop (and the name of that instance) in the design. "get design -worst_inst_drop …" valid after static/dynamic analysis. -limit <N> : displays the top N violations	
			minTW -limit <N>	get design -worst_inst_drop -dvd_type minTW : returns the worst minTW drop (and the name of that instance) in the design. -limit <N> : displays the top N violations	
			maxTW -limit <N>	get design -worst_inst_drop -dvd_type maxTW : returns the worst maxTW drop (and the name of that instance) in the design -limit <N> : displays the top N violations	
			avgTW -limit <N>	get design -worst_inst_drop -dvd_type avgTW : returns the worst avgTW drop (and the name of that instance) in the design -limit <N> : displays the top N violations	
	-worst_wire_drop	?-type?	power	get design -worst_wire_drop -type power : returns the power-net with worst wire drop and the worst wire drop value. "get design -worst_wire_drop …" valid after static/dynamic analysis	
			ground	get design -worst_wire_drop -type ground : returns the ground-net with worst wire drop and the worst wire drop value	
		?-net?	<netname>	get design -worst_wire_drop -net net1 : returns the worst wire drop of net1	
	-worst_em	?-net?	<netname>	get design -worst_em -net net1 : returns the EM% of net net1, gives error if net1 has no EM%. "get design -worst_em …" valid after static/dynamic analysis	
		?-type?	power	get design -worst_em -type power : returns the power net with worst EM% of all the power nets and the EM% value	
			ground	get design -worst_em -type ground : returns the ground net with first EM% of all the ground nets and the EM% value	
	-stats	?-count?	instances	get design -stats -count instances : returns the instance count of the design	print statistics
			nets	get design -stats -count nets : returns the net count of the design	
			memories	get design -stats -count memories : returns the count of memories the design	

	-mcyc_switching	-inst_file <file>	-o <out_file>	get design -mcyc_switching : returns switching time and switching state information of all instances, or a specified instance switching list, in dynamic simulation	
			blocks	get design -stats -count blocks : returns the instance count of sub-blocks the design	
	-layers	-used -type [metal| via] -o <output_file>		get design -layers -used : dumps layers that are actually present in the design (DEF/LEF files). Otherwise, the command dumps *all* metal/via layers from the Tech file	
	-spef_coverage			get design -spef_coverage provides a summary of SPEF parameters and their instance-wise coverage.	
	-sta_coverage			get design -sta_coverage provides a summary of STA parameters and their instance wise-coverage.	
	-vcd_coverage			get design -vcd_coverage provides a summary of VCD parameters and their instance wise-coverage.	
	-ipf_coverage			get design -ipf_coverage provides a summary of IPF parameters and their instance wise-coverage.	

get die

Command	Expression/ Target object	Options	Additional Options	Allowed Arguments	Description
get die	3D-IC dies	* -glob			returns a list of all dies in 3D-IC designs
	current_die	-current_die			returns the current die name dispalyed in the GUI
	bottom die	-bottom_most_die			returns the bottom die

get switch Command

Command	Expression/ Target object	Options	Additional Options	Allowed Arguments	Description	Overlap with existing commands
get switch	**<Target switch-name>**	-int_voltage			get switch sw1 -int_voltage : returns internal node voltage	
		-voltage_drop			get switch sw1 -voltage_drop : returns voltage-drop across the switch	
		-current			get switch sw1 -current : returns current through the switch	
		-net_pair			get switch sw1 -net_pair : returns external and internal net-pair in format VDD_NET1:VDD_NET2	
		-turn_on_time			get switch sw1 -turn_on_time : returns switch turn-on time	

get shorts Command

Command	Expression/ Target object	Options	Additional Options	Allowed Arguments	Description	Overlap with existing commands
get shorts	-net <net_name>	-layer <layername>	-bbox <llx> <lly> <urx> <ury>	-out_file <filename>	get shorts <options> returns all shorts present in the specified bbox. The locations reported are the mid points of the short	

get viamodel Command

Command	Expression/ Target object	Options	Additional Options	Allowed Arguments	Description	Overlap with existing commands
get viamodel	**<Target viamodel-name>**	-height			get viamodel via12 -height : returns height of the bounding-box	
		-width			get viamodel via12 -width : returns width of the bounding-box	
		-detail_bbox			get viamodel via12 -detail_bbox : returns details of bounding-box of each metal and cut layer for the via-model	

get instofcell

Command	Expression/ Target object	Options	Additional Options	Allowed Arguments	Description
get instofcell	<cell_type>	-o <output_file>			returns all instances of specified cell

get build Command

Command	Options	Options	Allowed Arguments	Description	Overlap with existing commands
get build	-name			get build -name: returns REDHAWK	
	-version			get build -version: returns REDHAWK version (for example, 9.2.p2)	
	-platform			get build -platform: returns platform (for example, Linux i686 2.4.21-47.ELsmp)	
	-date			get build -date: returns timestamp of the build	
				get build: returns all the above	

get current

Command	Expression/ Target object	Options	Additional Options	Allowed Arguments	Description
get current		-layer <via_cut_layer> -bbox <llx lly urx ury> -space <spacing_um> -o <output_file> (default: *.apache/.viacurrent*)			returns the sum of currents per via array in an output file, and displays the sum of the via currents per domain [-space : specifies the maximum distance between via cuts in order to consider the cuts as one and sum their currents. If the spacing is larger than specified, the two cut vias are treated as separate and currents are *not* summed. This option is used to find the original via array. Default is 0 (no cuts are summed).]

get layer

Command	Expression/ Target object	Options	Additional Options	Allowed Arguments	Description
get layer	<layername>	-location <x y> -voltage -current -hierarchy [1|0]	-out_file <filename>	? (-mode [RMS|AVG|PEAK] ?	returns the voltage or current at the specified layer and location. With -hierarchy 1, voltages/currents inside CMM cells are queried. -mode returns EM analysis mode.